

Capture the Flag 3: Data-Driven UI

Goal: Separate application data from the UI and handle multiple pieces of data in a usable interface.

🚩 Flag 0 — Create Your Feature Branch

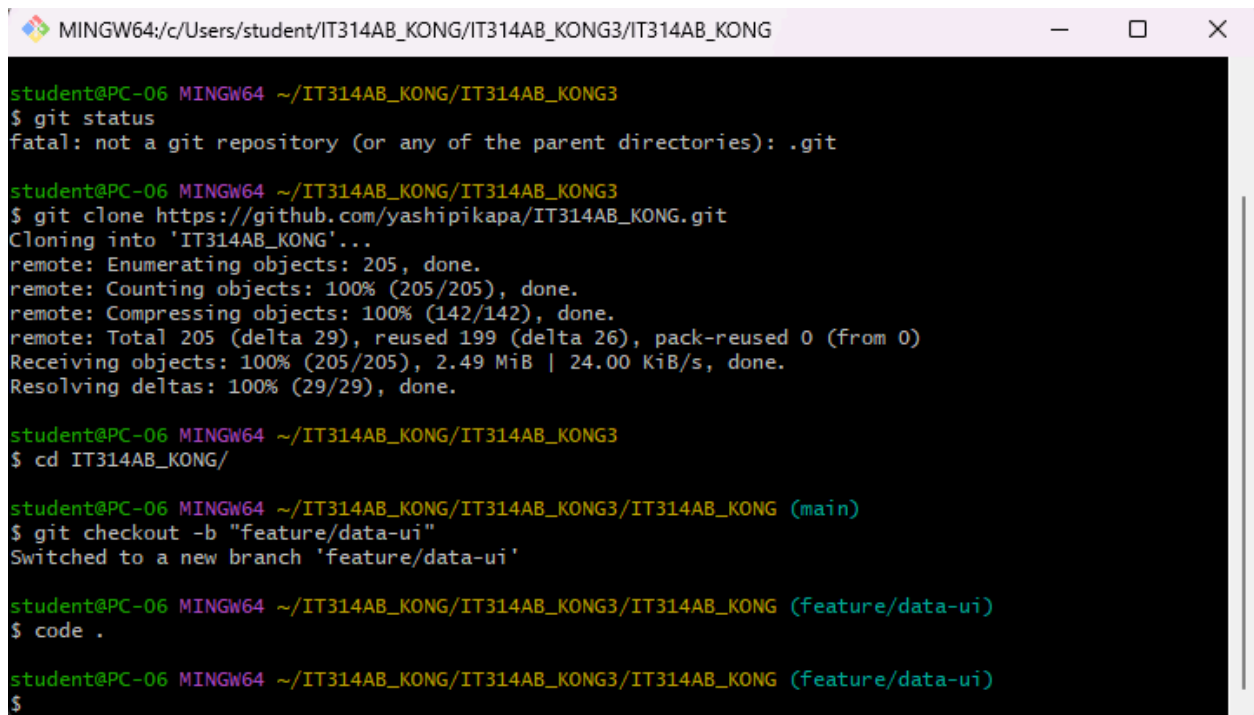
Objective: Create a feature branch before modifying your application.

Create:

feature/data-ui

Screenshot:

- Terminal showing your current branch



```
MINGW64; c:/Users/student/IT314AB_KONG/IT314AB_KONG3/IT314AB_KONG
student@PC-06 MINGW64 ~/IT314AB_KONG/IT314AB_KONG3
$ git status
fatal: not a git repository (or any of the parent directories): .git

student@PC-06 MINGW64 ~/IT314AB_KONG/IT314AB_KONG3
$ git clone https://github.com/yashipikapa/IT314AB_KONG.git
Cloning into 'IT314AB_KONG'...
remote: Enumerating objects: 205, done.
remote: Counting objects: 100% (205/205), done.
remote: Compressing objects: 100% (142/142), done.
remote: Total 205 (delta 29), reused 199 (delta 26), pack-reused 0 (from 0)
Receiving objects: 100% (205/205), 2.49 MiB | 24.00 KiB/s, done.
Resolving deltas: 100% (29/29), done.

student@PC-06 MINGW64 ~/IT314AB_KONG/IT314AB_KONG3
$ cd IT314AB_KONG/

student@PC-06 MINGW64 ~/IT314AB_KONG/IT314AB_KONG3/IT314AB_KONG (main)
$ git checkout -b "feature/data-ui"
Switched to a new branch 'feature/data-ui'

student@PC-06 MINGW64 ~/IT314AB_KONG/IT314AB_KONG3/IT314AB_KONG (feature/data-ui)
$ code .

student@PC-06 MINGW64 ~/IT314AB_KONG/IT314AB_KONG3/IT314AB_KONG (feature/data-ui)
$
```



Flag 1 — Identify the Data

Objective: Identify the appropriate Dart data type for the information in your profile.

Research basic Dart data types and identify at least **four different types** that could be used in your application.

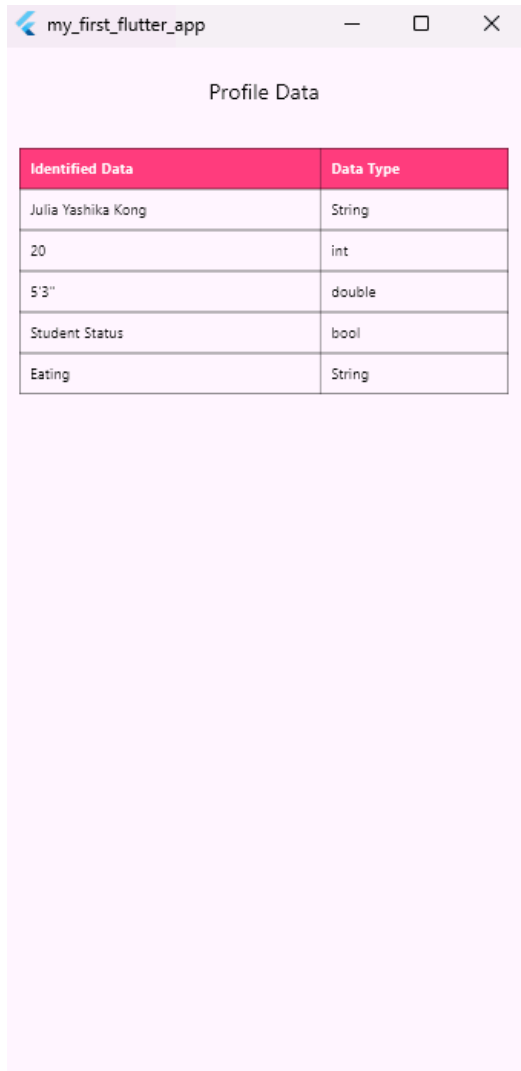
Examples:

String
int
double
bool

Create a table identifying the information and its appropriate data type.

Create:

- Your identified data
- The corresponding data types



The screenshot shows a Flutter application window with the title 'my_first_flutter_app'. Inside the window, there is a section titled 'Profile Data' which contains a table. The table has two columns: 'Identified Data' and 'Data Type'. The rows of the table are as follows:

Identified Data	Data Type
Julia Yashika Kong	String
20	int
5'3"	double
Student Status	bool
Eating	String

Flag 2 — Create Your Data

Objective: Create appropriately typed variables for your profile.

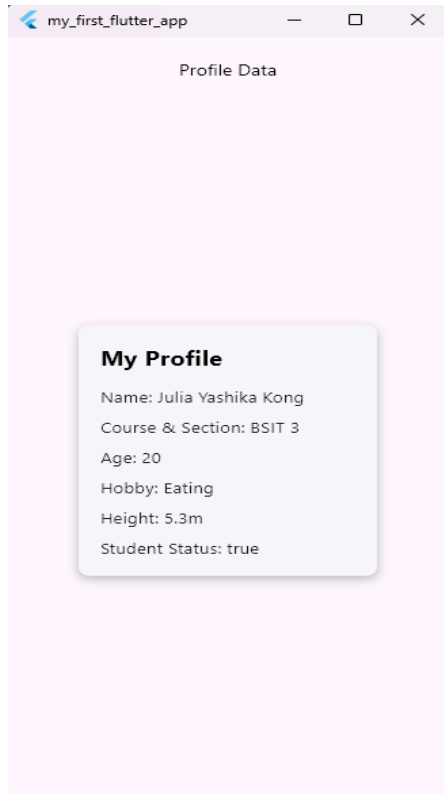
Create variables for at least:

- Name
- Course & Section
- Age
- Hobby
- Height
- Student status

Your data should exist **separately from your UI widgets**. (Declare variables separately from your UI, not directly inside your widgets)

Screenshot:

- Your variables
- Their data types
- Application still running



```
String name = "Julia Yashika Kong";  
String courseSection = "BSIT 3";  
int age = 20;  
String hobby = "Eating";  
double height = 5.3;  
bool isStudent = true;
```

Flag 3 — Connect Data to UI

Objective: Replace your hardcoded information with your variables.

Your profile should now display information using the variables you created.

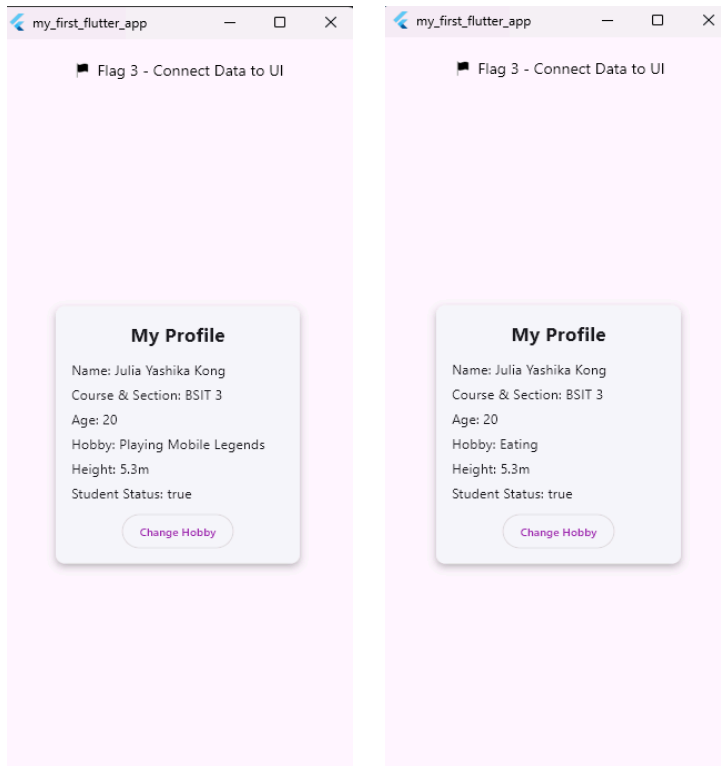
Test:

Change the value of one of your variables.

The UI should automatically display the new value.

Screenshot:

- Variables
- Widgets displaying the variables
- Updated application



🚩 Flag 4 — Make the Image Data-Driven

Objective: Make your profile image part of your data.

Create a variable containing the path to your profile image.

Use the variable when displaying the image.

Then change the variable to another registered image.

The image displayed by the application should change.

Screenshot:

- Image variable
- Image widget using the variable
- Updated application

```

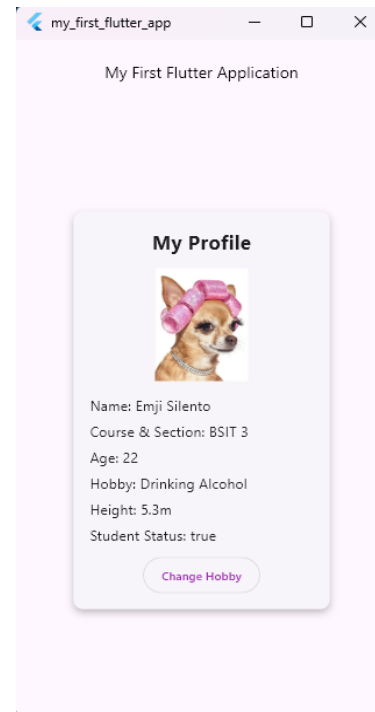
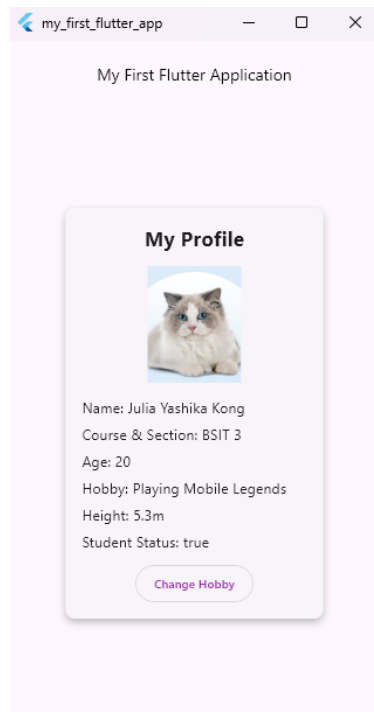
8 String ProfileImage = "assets/meow.png";
9 String appTitle = "My Profile";
10 String name = "Julia Yashika Kong";
11 String courseSection = "BSIT 3";
12 String birthdate = "March 14, 2003";
13 int age = 20;
14 String hobby = "Eating";
15 double height = 5.3;
16 bool isStudent = true;

```

```

8 String ProfileImage = "assets/cutie.jpg";
9 String appTitle = "My Profile";
10 String name = "Emji Silento";
11 String courseSection = "BSIT 3";
12 String birthdate = "February 24, 2004";
13 int age = 22;
14 String hobby = "Drinking Alcohol";
15 double height = 5.3;
16 bool isStudent = true;

```



🚩 Flag 5 — Five Profiles, One Design

Objective: Create at least **five different profile cards** using the same visual design.

Each profile must contain at least:

- Image
- Name
- Course & Section
- Age
- Hobby

The five profiles must contain different information.

Important

All five profiles should use the **same visual design**.

You are demonstrating that the same UI can represent different data.

Screenshot:

- Your 5 different profile data

```
8  String ProfileImage1 = "assets/cutie.jpg";
9  String name1 = "Julia Yashika Kong";
10 String courseSection1 = "BSIT 3";
11 int age1 = 20;
12 String hobby1 = "Eating";
13
14 String ProfileImage2 = "assets/meow.png";
15 String name2 = "Emji Silento";
16 String courseSection2 = "BSIT 3";
17 int age2 = 22;
18 String hobby2 = "Drinking Alcohol";
19
20 String ProfileImage3 = "assets/ha.jpg";
21 String name3 = "Leachim Dela Cerna";
22 String courseSection3 = "BSIT 3";
23 int age3 = 21;
24 String hobby3 = "Dancing";
25
26 String ProfileImage4 = "assets/pretty.png";
27 String name4 = "Karylle Racaza";
28 String courseSection4 = "BSIT 3";
29 int age4 = 21;
30 String hobby4 = "Dancing";
31
32 String ProfileImage5 = "assets/sexy.jpg";
33 String name5 = "Mav Sumalangit";
34 String courseSection5 = "BSIT 3";
35 int age5 = 21;
36 String hobby5 = "Singing & Dancing";
37
38 String appTitle = "My Profile";
```



Flag 6 — THE HIDDEN PROBLEM

You didn't see your UI?

Objective: Make all five profiles accessible to the user.

You just added five profile cards.

Now run your application.

Something is wrong.

Look carefully at your emulator.

Can you see all five profiles?

Maybe you can see the first one...

Maybe part of the second...

But where are the rest?

That's right.

You can't see them.

Your UI is larger than the available screen.

The profiles exist.

The data exists.

The widgets exist.

But the user can't access everything.

Your Mission:

Figure out how Flutter allows users to **scroll through content that is larger than the available screen**.

Research Flutter's **scrolling widgets** and determine which one is appropriate for your current layout.

You must make it possible to scroll through all five profiles.

Requirement:

The user must be able to:

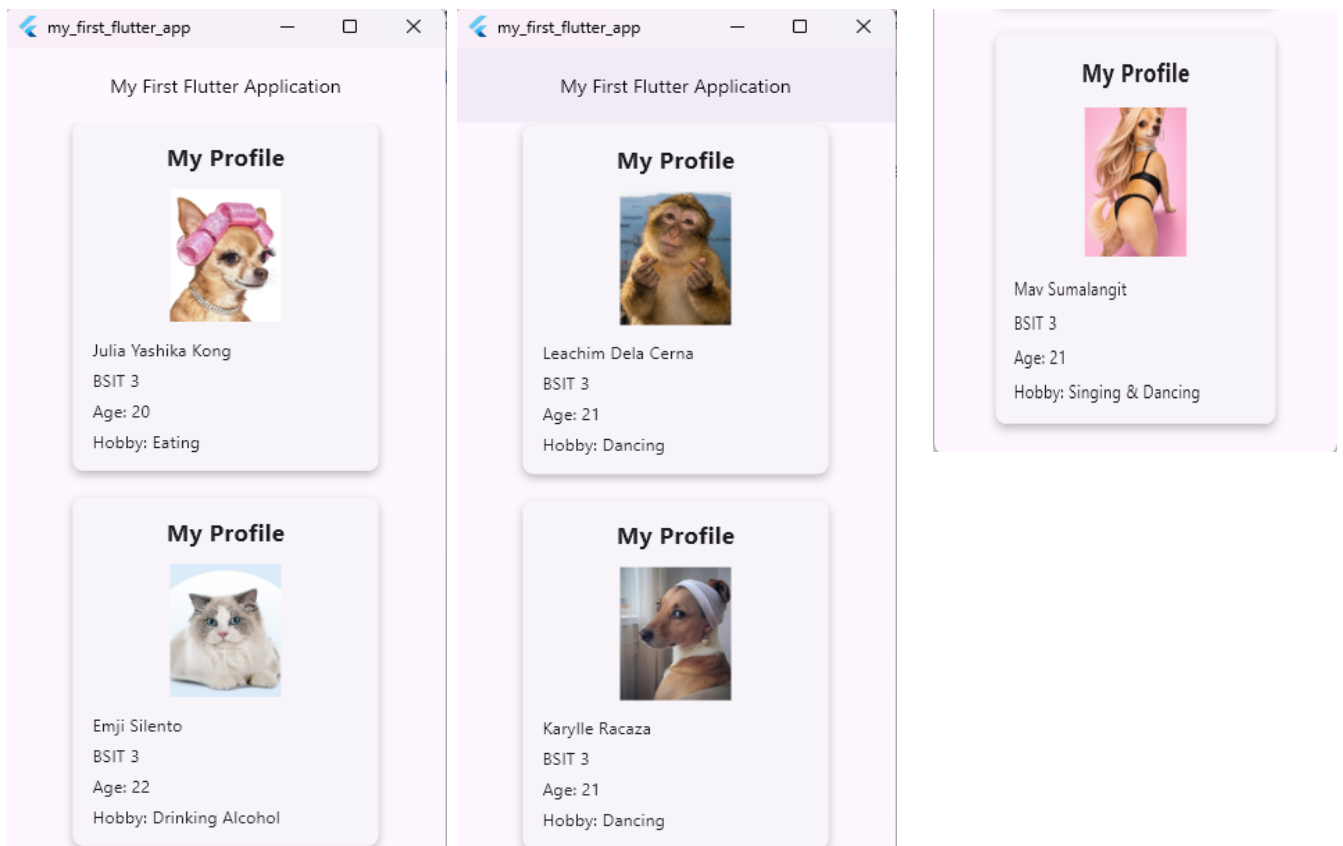
Scroll → **see Profile 1** → **Profile 2** → **Profile 3** → **Profile 4** → **Profile 5**

without changing the size of the screen.

Screenshot:

- The scrolling widget in your code
- Top portion of the application
- The rest of the portion of the application showing the later profiles

```
80  
81   body: SingleChildScrollView(  
82     child: Center(  
83       child: Column(  
84         children: [  
85  
86           // PROFILE 1  
87           Container(  
88             width: 220,  
89             margin: const EdgeInsets.only(bottom: 20),  
90             padding: const EdgeInsets.all(14),  
91
```



Flag 7 — Survive the Database

Objective: Prepare your UI for incomplete data.

Now imagine your profiles aren't manually created.

Someday, the data will come from a database.

And real data isn't always perfect.

Some information may be:

- Missing
- Empty
- `null`
- Not provided

Create at least **three profiles with missing information**.

For example:

Profile A

Name: Maria

Course: BSIT

Hobby: **missing**

Profile B

Name: Juan

Course: **missing**

Hobby: Gaming

Profile C

Name: **missing**

Course: MMA

Hobby: Drawing

Your application must **not crash** when information is missing.

Instead, provide an appropriate fallback.

Examples:

Hobby: Not provided

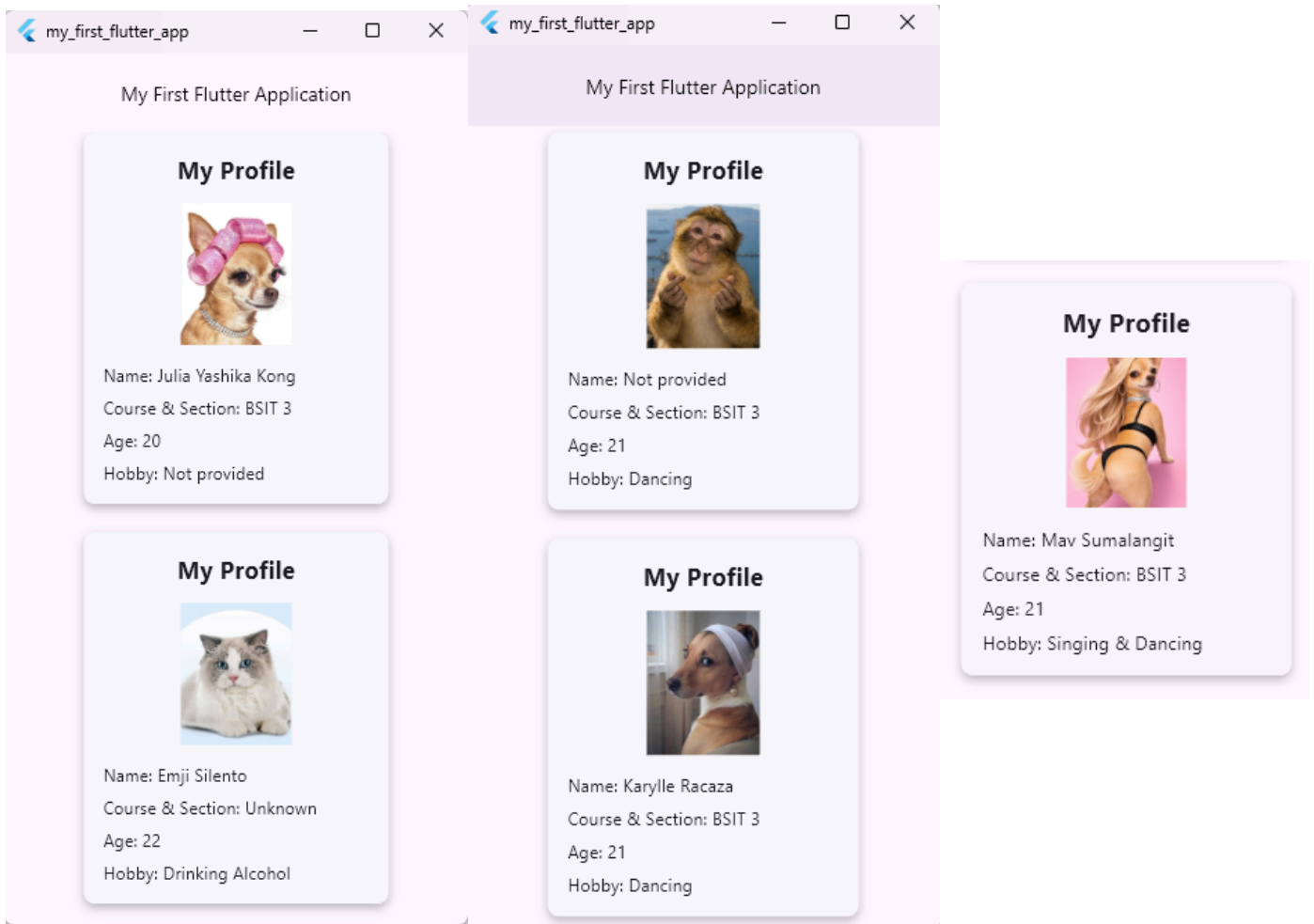
Course: Unknown

Challenge

Your placeholders should look intentional and clean.

Screenshot:

- Incomplete profile data
- Application displaying fallback values
- Application still functioning correctly



🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Data-Driven UI CTF.

Use an appropriate commit message following proper Git standards.

Branch:

feature/data-ui

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
PS C:\Users\student\IT314AB_KONG\IT314AB_KONG2\IT314AB_KONG\my_first_flutter_app> git commit -m "feat:Data-Driven UI Flutter Project"
[feature/data-ui 2dd544d] feat:Data-Driven UI Flutter Project
 7 files changed, 481 insertions(+), 222 deletions(-)
 create mode 100644 my_first_flutter_app/assets/cutie.jpg
 create mode 100644 my_first_flutter_app/assets/ha.jpg
 create mode 100644 my_first_flutter_app/assets/pretty.png
 create mode 100644 my_first_flutter_app/assets/sexy.jpg
PS C:\Users\student\IT314AB_KONG\IT314AB_KONG2\IT314AB_KONG\my_first_flutter_app> git push origin feature/data-ui
Enumerating objects: 19, done.
Counting objects: 100% (19/19), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (12/12), 705.25 KiB | 35.26 MiB/s, done.
Total 12 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 5 local objects.
remote:
remote: Create a pull request for 'feature/data-ui' on GitHub by visiting:
remote:   https://github.com/yashipikapa/IT314AB_KONG/pull/new/feature/data-ui
remote:
To https://github.com/yashipikapa/IT314AB_KONG.git
 * [new branch]   feature/data-ui -> feature/data-ui
PS C:\Users\student\IT314AB_KONG\IT314AB_KONG2\IT314AB_KONG\my_first_flutter_app> git status
On branch feature/data-ui
```